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Coalescence-induced droplet jumping on superhydrophobic surfaces with non-uniformly distributed micropillars

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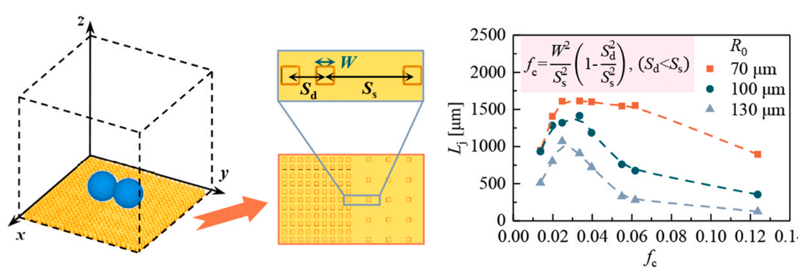
HIGHLIGHTS

- The droplet jumping on superhydrophobic non-uniform micropillar surfaces was studied.
- The effect of micropillar width, height, and spacing on droplet jumping was revealed.
- The solid fraction of the non-uniform micropillar surface was proposed.

GRAPHICAL ABSTRACT



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ABSTRACT

The phenomenon of droplet jumping induced by coalescence holds significant potential for application in diverse areas such as water collection, self-cleaning, and thermal management of electronic devices. In comparison to a flat surface, a superhydrophobic surface with structure can affect the interaction between the droplet and the surface, thereby affecting parameters such as the droplet jumping velocity. In this study, numerical simulation was used to examine the similarities and differences of coalescence-induced droplet jumping on superhydrophobic surfaces with uniform and non-uniform micropillars, as well as the effect of micropillar non-uniformity on droplet jumping. The results indicated that the non-uniform distribution of micropillars could lead to asymmetries in the shrinkage of the liquid bridge, the morphology of the droplet when it jumps off the surface, and the distribution of the high-pressure region at the bottom of the droplet. Additionally, it decreased the droplet's vertical velocity while substantially increasing the horizontal velocity. When the droplet radius was constant, decreasing the micropillar width or increasing the micropillar spacing on the sparse side was advantageous for the droplet jumping on a superhydrophobic surface with non-uniform micropillars. By defining the solid fraction on the micropillar surface with non-uniform distribution, the coupling effect of the width and spacing of the micropillars on the droplet jumping was thoroughly analyzed, and the optimal solid fraction for the best directional transport effect of droplets was determined. This study revealed the mechanism underlying the effect of surface micropillar structure on coalescence-induced droplet jumping, and the findings could guide the design of superhydrophobic surface microstructures, which was of great significance for the advancement of droplet jumping applications.

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1. Introduction

Coalescence-induced droplet jumping on superhydrophobic surfaces has significant potential in enhanced condensation [1–4], water harvesting [5–7], self-cleaning [8,9], and thermal management of electronic devices [10–12]. The phenomenon of droplet jumping is attributed to the interplay between the droplet and the surface, as well as the conversion of a portion of the surface energy that is released during droplet coalescence into kinetic energy that propels the droplet upward. In contrast to a flat surface, the presence of a structured superhydrophobic surface can exert an influence on the interplay between droplets and the surface. This, in turn, can have an impact on various parameters, including the direction and velocity of droplet jumping, thereby affecting the efficacy of coalescence-induced droplet jumping in practical applications.

In 2009, Boreyko et al. [13] presented the initial empirical findings of droplet jumping induced by coalescence on a superhydrophobic surface featuring carbon nanotubes and silicon micropillars. They conducted an experiment to determine the jumping velocity of droplets with varying radii, ranging from 8 μm to 160 μm. Their findings indicated that the observed jumping velocity adhered to the inertia-capillary scaling law. Afterward, there had been a significant amount of research conducted through experimentation, theoretical analysis, and computer simulations on the phenomenon of droplet jumping on structured superhydrophobic surfaces [14–27]. Xie et al. [14] constructed superhydrophobic surfaces featuring a single groove and ridge. They observed the coalescence and jumping phenomena of two droplets of equal dimensions on these surfaces and found that the impact of

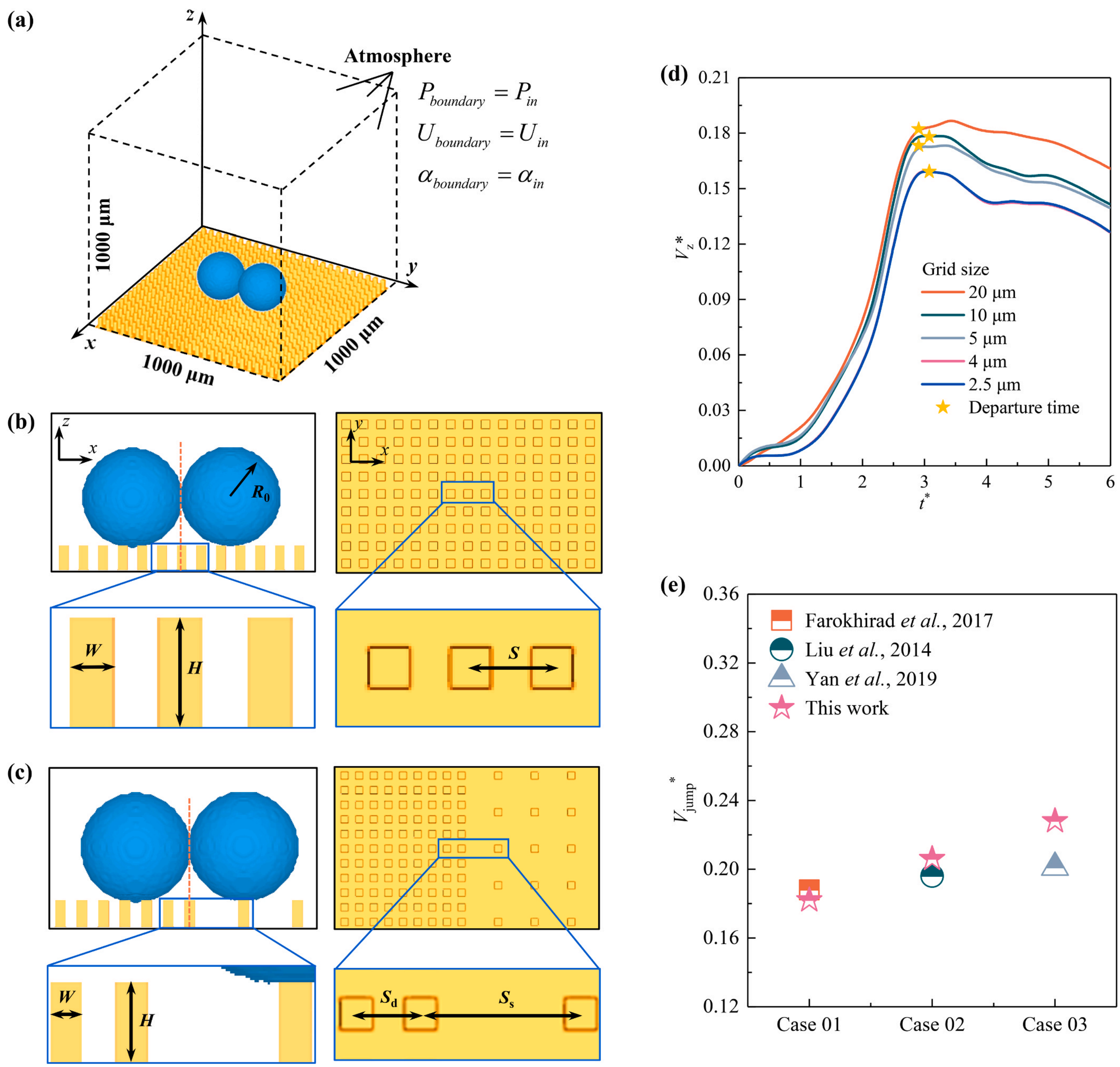

Fig. 1. Schematic diagram of the numerical simulation calculation domain, the grid independence test, and the results of the model validation. (a) The calculation domain and boundary conditions for the numerical simulation; (b)-(c) The side view of the initial state of the droplet and the top view of the distribution of micropillars on the superhydrophobic surface with uniform and non-uniform micropillars; (d) The grid independence test results; (e) The results of numerical model verification.

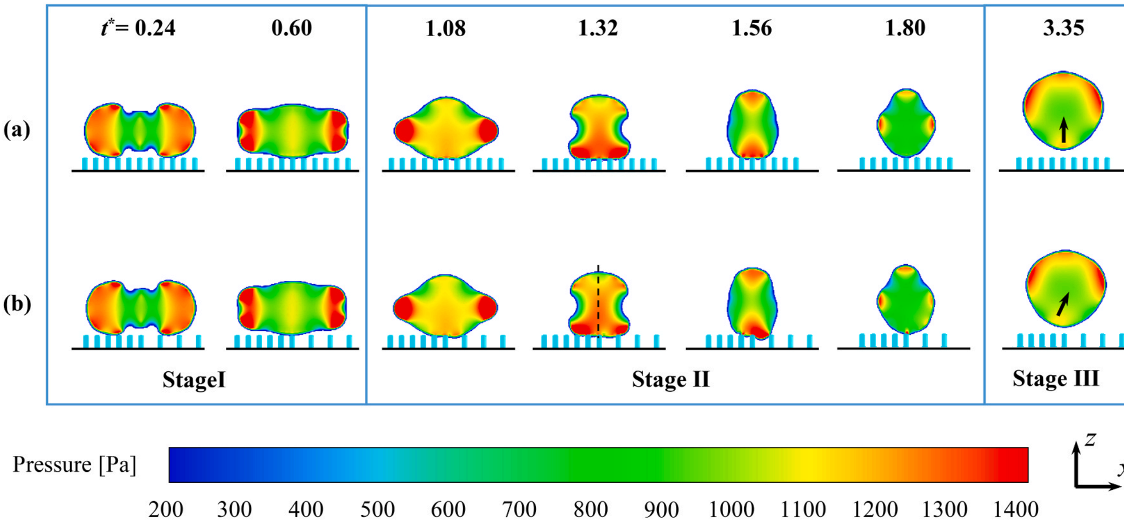

Fig. 2. Time evolution of the pressure inside the droplet during the coalescence-induced jumping. The width W and the height H of the micropillars are 20 μm and 50 μm, respectively. (a) The uniform micropillar surface and the micropillar spacing S is 40 μm; (b) The non-uniform micropillar surface and the micropillar spacing S_d and S_s are 40 μm and 70 μm, respectively. The initial droplet radius R_0 = 100 μm. Stage I is the formation and expansion of a liquid bridge between two droplets, Stage II is the shrinkage of the liquid bridge, and Stage III is the detachment of the droplet from the surface.

groove and ridge on droplet coalescence kinetics was altered in comparison to superhydrophobic flat surfaces, resulting in reduced viscous dissipation and coalescence time. In addition to a single structure, the formation of micropillar arrays on superhydrophobic surfaces also affects droplet dynamics. The study conducted by Farokhirad et al. [17] demonstrated that the droplet jumping velocity and height were amplified on a surface featuring a microstructure of a columnar array, in comparison to a flat surface. Additionally, the range of droplet contact angles that enable droplet jumping was expanded. Yin et al. [18] derived a theoretical model to determine the critical condition for droplet jumping on a micropillar surface and identified the optimal condition for achieving maximum jumping velocity. Furthermore, the height, width, and spacing of the micropillars also affected the droplet jumping height and velocity. The findings of Shi et al. [24] indicated that the height of conical posts and the spacing between them had a significant impact on the jumping height of droplets on a surface with uniformly distributed arrays of such micropillars. Specifically, their numerical results demonstrated that an increase in the height of the conical posts and a decrease in the spacing between them led to an increase in the droplet jumping height. The study conducted by Liu et al. [25] involved the simulation of droplets jumping on a surface featuring micropillars that were uniformly distributed. The findings of the study indicated that there existed an optimal micropillar spacing for constant droplet size. At this particular spacing, the droplet could attain the highest possible jumping height. Attarzadeh and Dolatabad's findings indicated that the micropillar width and spacing influenced the droplet volume immersed in the micropillar gap, which in turn influenced the droplet's jumping velocity [27]. The greater the droplet's volume immersion in the micropillar gap, the greater the viscous dissipation and the lower the droplet's jumping velocity.

Despite the considerable amount of research conducted on the phenomenon of coalescence-induced droplet jumping on surfaces featuring micropillars, certain limitations persist. First of all, the existing studies primarily concentrate on droplet jumping on surfaces featuring uniformly distributed micropillars. However, there exists a dearth of research about the shape evolution and velocity of droplets on surfaces that feature non-uniformly distributed micropillars. The non-uniform arrangement of micropillars can have an impact on the interfacial behavior of droplets and surfaces, leading to alterations in droplet morphology, pressure distribution, and volume of droplet immersion within the pillar gaps. The impact of the aforementioned parameters remains uncertain. Secondly, on the surface with non-uniform distributed micropillars, the hydrophobicity of the surface is weaker where the spacing of the micropillars is small. As a result, droplets exhibit spontaneous movement from the less dense side of the micropillars to the denser side [28,29], and the horizontal velocity of droplets is a crucial parameter in practical scenarios. Hence, it is imperative to investigate the pattern of velocity fluctuation in the horizontal direction while the droplet jumps off a micropillar-patterned surface. This work employs numerical simulation to compare the coalescence-induced droplet jumping on superhydrophobic surfaces with uniform and non-uniform distributed micropillars. And the impact of the non-uniformity of micropillar distribution on droplet shape evolution and velocity during the droplet jumping process is also investigated. The findings of this study could provide theoretical guidance for the practical application of droplet jumping in various fields such as water collection, self-cleaning, and thermal management of electronic devices.

2. Numerical methods

The interFoam multiphase flow solver, founded on the VOF (Volume of Fluids) model and implemented in the OpenFOAM software, was employed to conduct a simulation of the coalescence and jumping of droplets on the surface featuring micropillars. The fundamental governing equations comprise the continuity equation and the momentum equation:

$$\nabla\cdot\boldsymbol{u} = 0 \tag{1}$$

$$\frac{\partial(\rho\boldsymbol{u})}{\partial t} + \nabla\bullet(\rho\boldsymbol{u}\boldsymbol{u}) = \nabla\bullet\mu\left[\nabla\boldsymbol{u} + (\nabla\boldsymbol{u})^{\mathrm{T}}\right] + \rho g - \nabla p + \mathbf{F}_{\mathbf{SF}} \tag{2}$$

Where $\boldsymbol{u}$ represents the velocity vector, t denotes time, p stands for pressure, and g represents gravitational acceleration. The volumetric force term produced by the surface tension, $\mathbf{F}_{\mathbf{SF}}$, is determined using the CSF (continuum surface force) model: $\mathbf{F}_{\mathbf{SF}} = \sigma\kappa\nabla\alpha$. And $\kappa = -\nabla\bullet\boldsymbol{n}$ represents the curvature of the phase interface, $\boldsymbol{n} = \nabla\alpha/|\nabla\alpha|$ denotes the unit normal vector of the phase interface. The variables ρ and μ correspond to the average density and average viscosity, respectively:

$$\rho = \alpha_{\mathrm{w}}\rho_{\mathrm{w}} + \alpha_{\mathrm{a}}\rho_{\mathrm{a}} \tag{3}$$

$$\mu = \alpha_{\mathrm{w}}\mu_{\mathrm{w}} + \alpha_{\mathrm{a}}\mu_{\mathrm{a}} \tag{4}$$

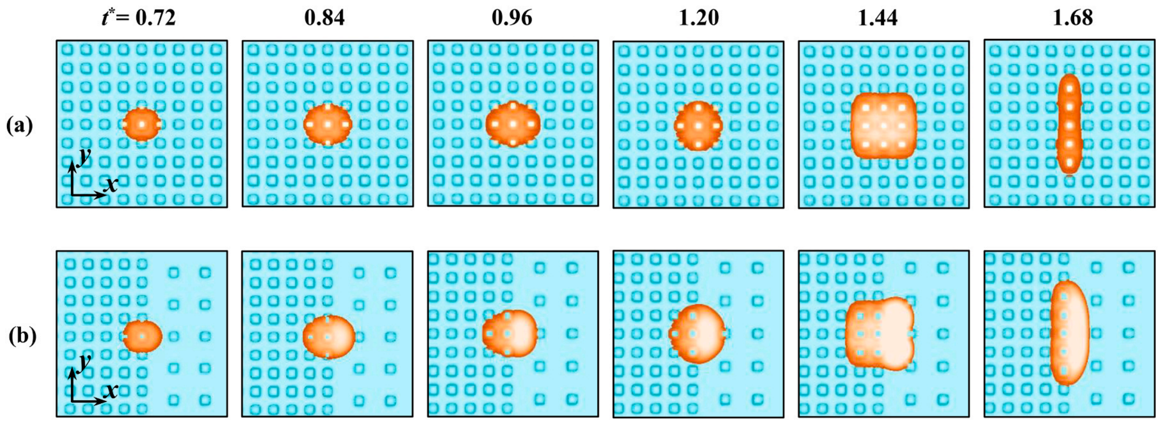

Fig. 3. The evolution of the solid-liquid contact section at the top of the micropillar overtime during the shrinkage of the liquid bridge. (a) The uniform micropillar surface, micropillar width W= 20 μm, height H= 50 μm, spacing S= 40 μm; (b) The non-uniform micropillar surface, micropillar width W= 20 μm, height H= 50 μm, the spacing of the dense side S_d= 40 μm, and the spacing of the sparse side S_s = 70 μm.

Where α is the phase fraction, and the subscripts w and a represent water and air, respectively.

The droplet velocity **V** is calculated by the following formula:

$$\mathbf{V} = \frac{\int_{\Omega} \alpha(\boldsymbol{u})\mathrm{d}\Omega}{\int_{\Omega} \alpha \mathrm{d}\Omega} \tag{5}$$

Where Ω represents the computational domain size.

Fig. 1(a) depicted the computational domain utilized in the numerical simulation, with dimensions of 1000 μm for its length, width, and height. In the simulation, the surface featuring micropillars was considered a no-slip wall, while the remaining boundaries were defined as pressure outlet boundary conditions. At the onset, two spherical droplets possessing a radius of R_0 were in contact and situated atop the micropillar. The cross-sectional shape of the micropillar on the solid wall was depicted in Fig. 1(b)-(c), wherein it was observed to be a square. The dimensions of the micropillar, namely the width, height, and spacing, were denoted by W, H, and S, respectively. The point of tangency between the two droplets was on the central axis of a given micropillar on the surface featuring uniformly distributed micropillars. And the surface featuring non-uniformly distributed micropillars was confined by the centerline of the calculation domain (y = 500 μm), and the region was categorized into areas of high and low density. The distribution of micropillars was uniform across each region. The micropillar spacing on the dense side and the sparse side were denoted as S_d and S_s, respectively. The point of tangency between the two droplets lay on the midline of the micropillar separating the regions of high and low density. The grid independence test results were presented in Fig. 1(d). To optimize computational efficiency without compromising precision, a grid size of 4 μm was chosen for this study. The results of numerical model verification were illustrated in Fig. 1(e) and the results indicated that the dimensionless center-of-mass jumping velocity V_{jump}* was close to the findings reported in the literature [17,30,31], thereby substantiating the dependability of the simulation.

3. Results and discussions

3.1. Comparison of the coalescence-induced droplet jumping on surfaces with uniform and non-uniform micropillars

The pressure evolution within the droplet during the jumping process on both the uniform micropillar surface W20H50S40 and the non-uniform micropillar surface W20H50S40–70 were depicted in Fig. 2. The dimensions of the micropillars on the surfaces of W20H50S40 and W20H50S40–70 were identical in terms of width (W=20 μm) and height (H=50 μm). However, the spacing (S) between the micropillars differed between the two surfaces. For the surface W20H50S40–70, the spacing between the micropillars located on the sparse and dense sides of the droplet's tangent line was asymmetrical, measuring 40 μm and 70 μm, respectively. As shown in Fig. 2, the process of droplet coalescence and jumping on a micropillar surface could be segmented into three stages including (I) the formation and expansion of a liquid bridge between two droplets, (II) the shrinkage of the liquid bridge, and (III) the detachment of the droplet from the surface. During the stage of liquid bridge formation and expansion, the interfacial tension of two contacting droplets resulted in the creation of a liquid bridge, which subsequently underwent rapid expansion and ultimately came into contact with the wall. At this stage, the internal pressure distribution of the droplet on the two surfaces exhibited nearly identical characteristics, evincing high pressure on both ends and low pressure in the center, with minimal influence from the micropillar distribution. During the stage of shrinkage in the liquid bridge that followed, it could be seen from Fig. 2(a) and (b) that the non-uniform arrangement of micropillars had a discernible effect. The non-uniform micropillar surface induced asymmetry in the shrinkage of the liquid bridge and the distribution of the high-pressure area at the bottom of the droplet on the sparse side. This was attributed to the immersion of the droplet's bottom into the micropillar gap (t*=1.32, 1.56). At t*= 1.32, a pressure differential existed between the left and right sides of the droplet, leading to an unbalanced horizontal force and consequent acceleration. This acceleration could be calculated by:

$$a = \frac{3\Delta p A_p}{8\pi R_0^3 \rho_w} \tag{6}$$

a= 2.8 m/s^2 could be obtained using the average pressure difference on the left and right sides of the droplet denoted as Δp and the interface area denoted as A_p. Eventually, the jumping direction of the droplet was not perpendicular to the substrate (Stage III) and the droplet could be transported a certain distance in the horizontal direction after leaving the surface.

The area of the gas-liquid interface decreased as the droplet coalesced on the surface with micropillars and surface energy was released. The surface energy ΔE_s released by the droplet at a particular instant was the difference between the surface energy at the initial instant and the surface energy at this moment. A portion of the released surface energy would be converted into the kinetic energy of the droplet, causing it to jump off the surface, while the rest was consumed by viscous dissipation and solid-liquid adhesion. Previous studies have shown that the asymmetry of the solid-liquid interaction would have an impact on the ratio of the surface energy released to the kinetic energy of

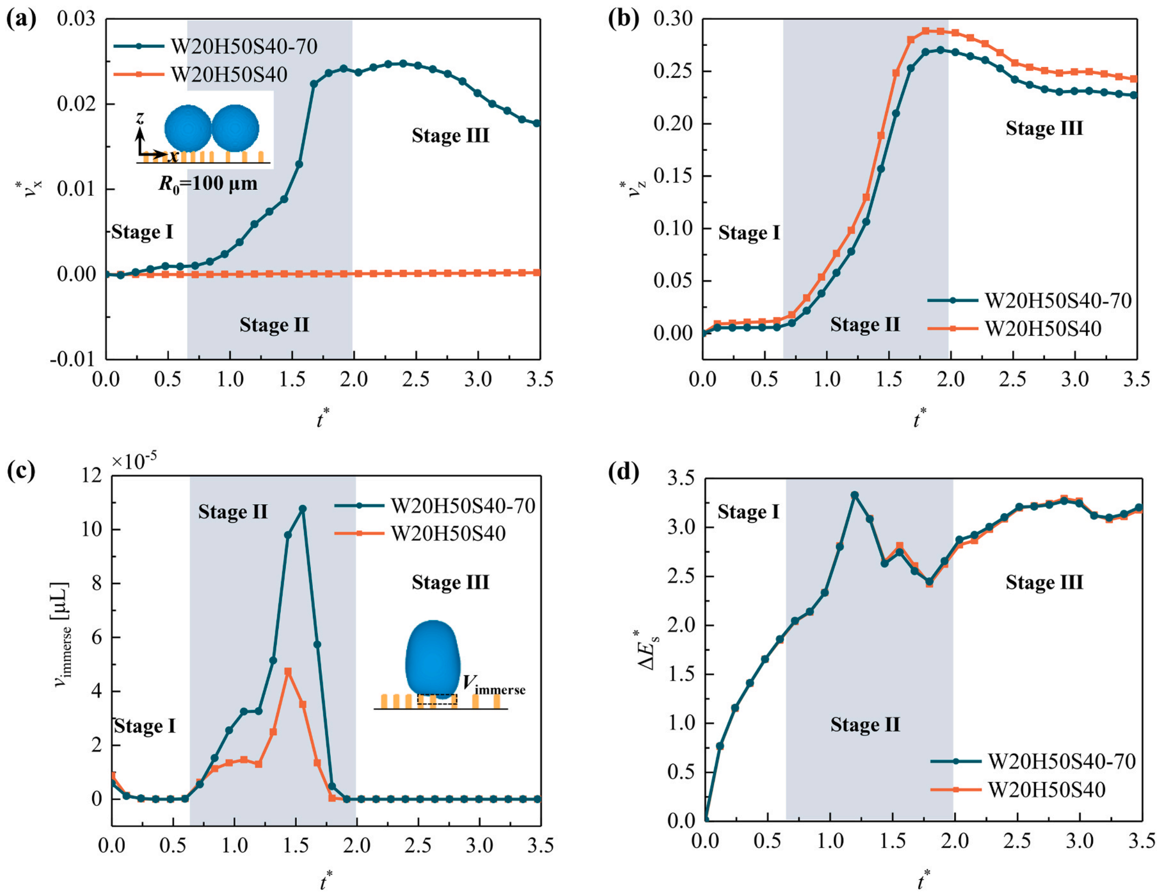

Fig. 4. On the uniform micropillar surface W20H50S40 and the non-uniform micropillar surface W20H50S40–70, during the coalescence-induced droplet jumping process (a)-(b) the variation of the dimensionless center-of-mass velocity in the horizontal and vertical direction with time; (c) the variation of the liquid volume immersed in the gap between the micropillars with time; (d) the variation of the dimensionless released surface energy with time.

jumping, which would subsequently have an impact on the droplet velocity and other parameters [32–37]. The evolution of the solid-liquid contact section at $z = 50$ μm over time on the surface with uniform and non-uniform micropillars during the shrinkage of the liquid bridge was shown in Fig. 3. Comparing Fig. 3(a) and (b), it could be seen that the length of the solid-liquid contact surface in the x direction increased first and then decreased with time, and the length in the y direction increased gradually in both cases. On the surface with non-uniformly distributed micropillars, there was more liquid immersed in the gap of the sparse side, and the solid-liquid contact section was asymmetrical in the x direction.

Consistent with previous research [3,38], the droplet velocity **V** and time t were dimensionless with the capillary-inertial velocity V_{ic} ($= \sqrt{\sigma/\rho R_e}$) and the capillary-inertial time t_{ic} ($= \sqrt{\rho R_e^3/\sigma}$), respectively: $\mathbf{V}^*=\mathbf{V}/V_{ic}$, $t^*= t/t_{ic}$. And the released surface energy was dimensionless by σR_e^2: $\Delta E_s^* = \Delta E_s/\sigma R_e^2$. Dimensionless velocity $\mathbf{V}^*$, dimensionless time t^*, and dimensionless released surface energy ΔE_s^* were used to characterize the coalescence and jumping behavior of droplets on surfaces with micropillars. The equivalent radius after two droplets coalesce into a single droplet is:

$$R_e = \left[2R_0^3\left(1 - \frac{(1+\cos\theta)^2(2-\cos\theta)}{4}\right)\right]^{\frac{1}{3}} \quad (7)$$

Where θ was the contact angle of the droplet on the surface.

Fig. 4(a) and (b) illustrated the variation of the dimensionless center-of-mass velocity in the horizontal (x) and vertical (z) directions with time during the droplet jumping on the uniform micropillar surface W20H50S40 and the non-uniform micropillar surface W20H50S40–70. During the stage of liquid bridge formation and expansion (Stage I), the disparity in horizontal velocity of the droplet on both surfaces was negligible. During the stage of liquid bridge shrinkage, the presence of non-uniformly distributed micropillars could lead to a notable enhancement in the horizontal velocity of the droplet, and this effect caused the droplet to migrate from the region of higher micropillar density towards the region of lower micropillar density, which was attributed to the asymmetric distribution of pressure within the droplet, causing a horizontal pressure gradient and subsequently generating a net force from the region of higher density to that of lower density. Simultaneously, the non-uniform distribution of micropillars would result in a reduction of the droplet velocity in the vertical direction. This was due to the increased spacing between the micropillars led to a decrease in the reaction force of the solid wall on the droplet. Although the horizontal velocity was smaller than the vertical velocity on the non-uniform micropillar surface, droplets' horizontal velocity and transport distance were still substantially greater than those on the uniform micropillar surface (Section S1 of the Supplementary material). Fig. 4(c) depicted the time-dependent changes in the liquid volume immersed in the gap between the micropillars on the two surfaces. On the non-uniform micropillar surface, due to the large spacing between the sparse side micropillars, the liquid volume immersed in the micropillar gap increased, especially in the stage of liquid bridge shrinkage (Stage II). The change of the dimensionless released surface energy with time was shown in Fig. 4(d). The figure illustrated that the uniformity of the distribution of micropillars had minimal impact on the dimensionless released surface energy ΔE_s^*. It should be noted that when the initial droplet position and the size ratio of the two droplets were altered, the

Table 1
Structural parameters of the surface with different non-uniform micropillars.

Surface	W μm	H μm	S_d μm	S_s μm	θ_d °	θ_s °
W10H50S40–70	10	50	40	70	172.6	175.8
W12H50S40–70	12	50	40	70	171.1	174.9
W20H50S40–70	20	50	40	70	165.1	171.5
W30H50S40–70	30	50	40	70	157.6	167.3
W20H20S40–70	20	20	40	70	165.1	171.5
W20H80S40–70	20	80	40	70	165.1	171.5
W20H50S40–60	20	50	40	60	165.1	170.1
W20H50S40–90	20	50	40	90	165.1	173.4
W20H50S40–100	20	50	40	100	165.1	174.1
W20H50S40–120	20	50	40	120	165.1	175.1

horizontal velocity, vertical velocity, liquid volume immersed in the gap between micropillars, and released surface energy were all affected (Section S2 and S3 of the Supplementary material). When the morphology of the micropillar altered, the droplet also had a certain horizontal velocity when jumping off the surface with non-uniform micropillars (Section S4 of the Supplementary material). Overall, compared with the uniform micropillar surface, the non-uniform micropillar surface was more conducive to the increase of the droplet velocity in the horizontal direction.

3.2. Influence of surface structure parameters with non-uniform micropillars on droplet jumping

The droplet jumping behavior exhibited a significant reliance on the interaction between the solid and liquid phases, as evidenced by the preceding observations. The solid-liquid interaction was primarily influenced by the structural parameters of micropillar width, height, and spacing of non-uniform micropillar surfaces. The contact angle of a Cassie−Baxter state droplet on a non-uniform micropillar surface was:

$$\cos\theta_d = f_d(\cos\theta_f + 1) - 1 \tag{8}$$

$$\cos\theta_s = f_s(\cos\theta_f + 1) - 1 \tag{9}$$

Where $f_d = W^2/S_d^2$ and $f_s = W^2/S_s^2$ represent the solid fractions of droplets located on the top of micropillars situated on the dense and sparse sides of a non-uniform micropillar surface in the Cassie-Baxter state, respectively. θ_f was the contact angle of a droplet on a smooth surface, which was 150° in this study, and the change of θ_f had little effect on droplet jumping (Section S5 of the Supplementary material). To investigate the impact of micropillar width, height, and spacing on coalescence-induced jumping, the droplet jumping phenomenon was analyzed on the surface of ten distinct micropillar configurations, as illustrated in Table 1.

The effect of micropillar width on droplet jumping behavior on the non-uniform micropillar surface was demonstrated in Fig. 5. The height of micropillars on the three surfaces was H= 50 μm, the spacing of micropillars was S_d= 40 μm and S_s = 70 μm, but the widths of micropillars W were 10 μm, 20 μm, and 30 μm, respectively. The variation of

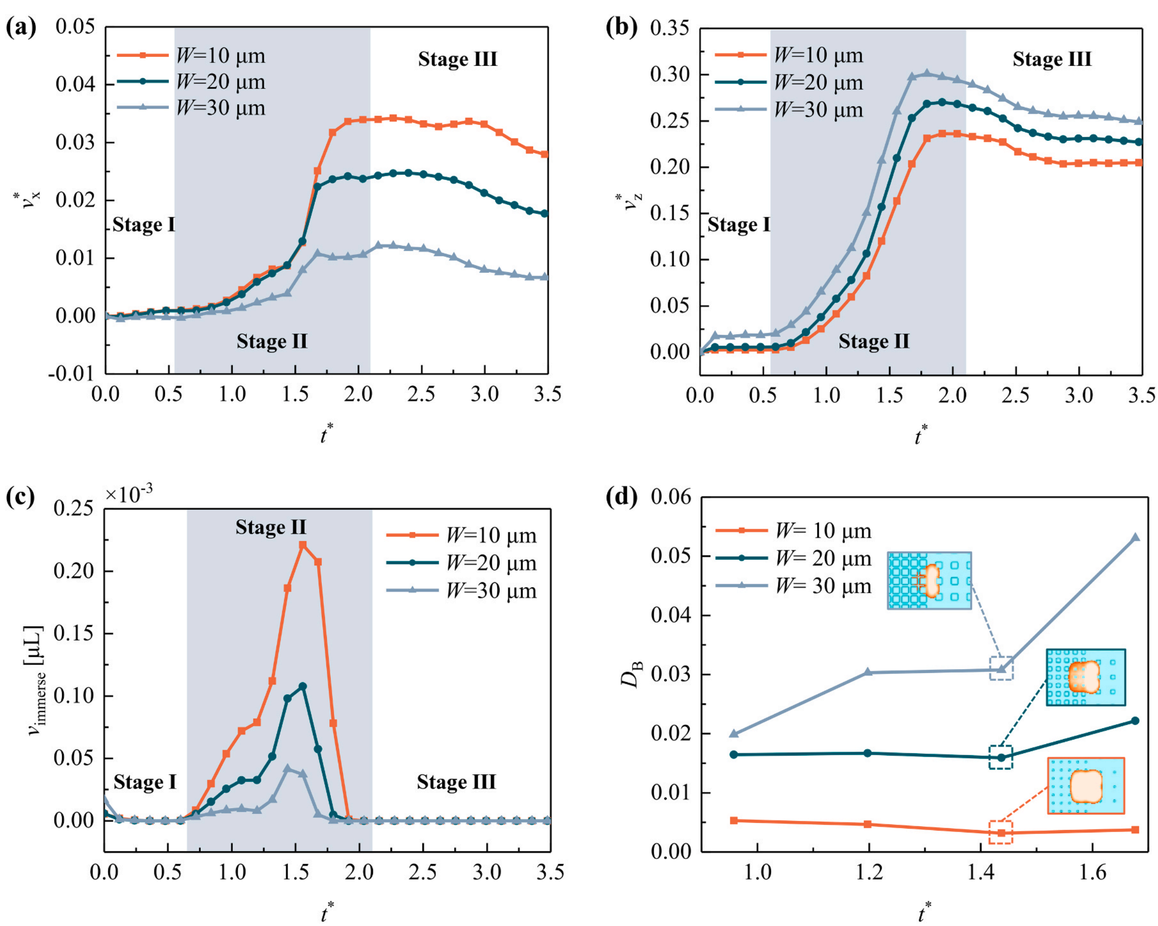

Fig. 5. Effect of micropillar width on droplet jumping on the non-uniform micropillar surface. (a)-(b) The variation of dimensionless center-of-mass velocity in the horizontal and vertical direction changes with time; (c) The variation of the liquid volume immersed into the micropillar gap with time; (d) The variation of the symmetry degree of the solid-liquid contact surface with time. On the three surfaces, the micropillar height is H= 50 μm, the micropillar spacing on the dense side is S_d= 40 μm and the micropillar spacing on the sparse side is S_s = 70 μm.

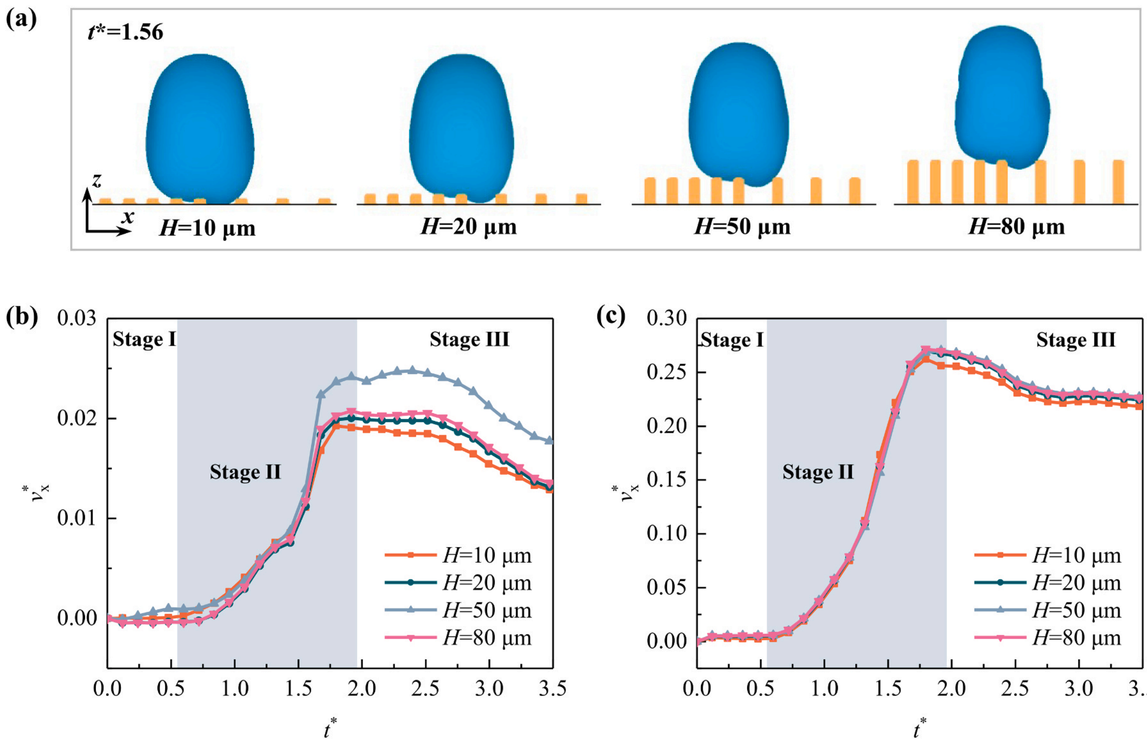

Fig. 6. Effect of micropillar height on droplet jumping on the non-uniform micropillar surface. (a) The morphology of the droplet when it moves to the bottom; (b)-(c) The variation of dimensionless center-of-mass velocity in the horizontal and vertical directions changes with time. On the four surfaces, the micropillar width is W= 20 μm, the micropillar spacing on the dense side is S_d= 40 μm and the micropillar spacing on the sparse side is S_s = 70 μm.

the dimensionless center-of-mass velocity in the horizontal direction with time was presented in Fig. 5(a). During the stage of liquid bridge formation and expansion (Stage I), the horizontal velocities on the three surfaces were all close to zero. In the subsequent stage of liquid bridge shrinkage (Stage II), as depicted in Fig. 5(a) and (c), it was evident that the horizontal velocity of the droplet gradually increased as the liquid volume immersed in the micropillar gap increased. This phenomenon could be attributed to the asymmetric solid-liquid interaction occurring on the surface of the non-uniform micropillar. Overall, the horizontal velocity decreased as the micropillar width increased, and the horizontal velocity of the droplet when it detached from the non-uniform micropillar surface with a width of 10 μm was approximately three times that of the surface with a width of 30 μm. After the droplet jumped off the surface (Stage III), the solid-liquid interaction disappeared, and the horizontal velocity remained unchanged at first, but as the droplet oscillated upward in the air, the energy was further dissipated, leading to a gradual decrease in the horizontal velocity. However, although the smaller width of the micropillar surface was beneficial to the increase of the horizontal velocity, more liquid immersion into the micropillar gap resulted in a decline of the droplet velocity in the vertical direction, as depicted in Fig. 5(b). The solid-liquid contact cross-sections during the droplet jumping process on three surfaces were subjected to further processing using the gray histogram. The asymmetry of the solid-liquid contact surface between the dense side and the sparse side of the micropillars was characterized by obtaining the Bhattacharyya distance D_B [39]. The larger the D_B was, the more asymmetric the solid-liquid contact surface between the two sides was. The change of the symmetry degree of the solid-liquid contact surface (i.e., the Bhattacharyya distance D_B) on the three surfaces with time was shown in Fig. 5(d). The asymmetry of the solid-liquid interface increased with the increase of the micropillar width. From Fig. 5(d), it could be deduced that the horizontal velocity was primarily influenced by the immersion of the liquid into the gap between the micropillars, rather than the asymmetry of the solid-liquid contact surface. In the case of a micropillar surface with a width of 30 μm, despite the solid-liquid contact surface exhibiting the greatest degree of asymmetry, the droplet was impeded from immersing into the micropillar gap due to the increased width of the micropillar. Consequently, the velocity of the liquid droplet on the surface in the horizontal direction remained comparatively smaller.

The effect of micropillar height on droplet jumping on the non-uniform micropillar surface was demonstrated in Fig. 6. For surfaces with micropillar width W= 20 μm, dense side micropillar spacing S_d= 40 μm, sparse side micropillar spacing S_s = 70 μm, and micropillar height H was 10 μm, 20 μm, 50 μm, and 80 μm, respectively, the micropillar height had a negligible effect on the droplet jumping, regardless of whether the droplet made contact with the plane at the bottom of the micropillar (Fig. 6(a)). The findings illustrated in Fig. 6(b) and (c) indicated that the horizontal velocity of the droplet exhibited a non-monotonic trend as the micropillar height increased, initially increasing and subsequently decreasing. However, it was noteworthy that the difference in the horizontal velocity of the droplet on the three surfaces was not significant, and changes in micropillar height also had little effect on vertical velocity.

Another structural parameter that determined the droplet jumping performance on non-uniform micropillar surfaces was the micropillar spacing. When studying the effect of micropillar spacing on droplet jumping, the micropillar spacing on the dense side S_d was maintained at 40 μm, while the micropillar spacing on the sparse side gradually increased. Fig. 7 depicted the dimensionless center-of-mass velocity of the droplet in the horizontal and vertical directions, the liquid volume immersed into the micropillar gap, and the symmetry degree of the solid-liquid contact surface over time on three surfaces with different micropillar spacings. Fig. 7(a) and (b) indicated that as the distance between the micropillars on the sparse side increased, there was a gradual increase in the horizontal velocity of the droplet. However, the vertical velocity remained relatively the same. Upon the pressure distribution and velocity vector diagrams inside the droplet of the sparse side micropillar spacing of S_s = 70 μm and 100 μm, as depicted in Fig. 7

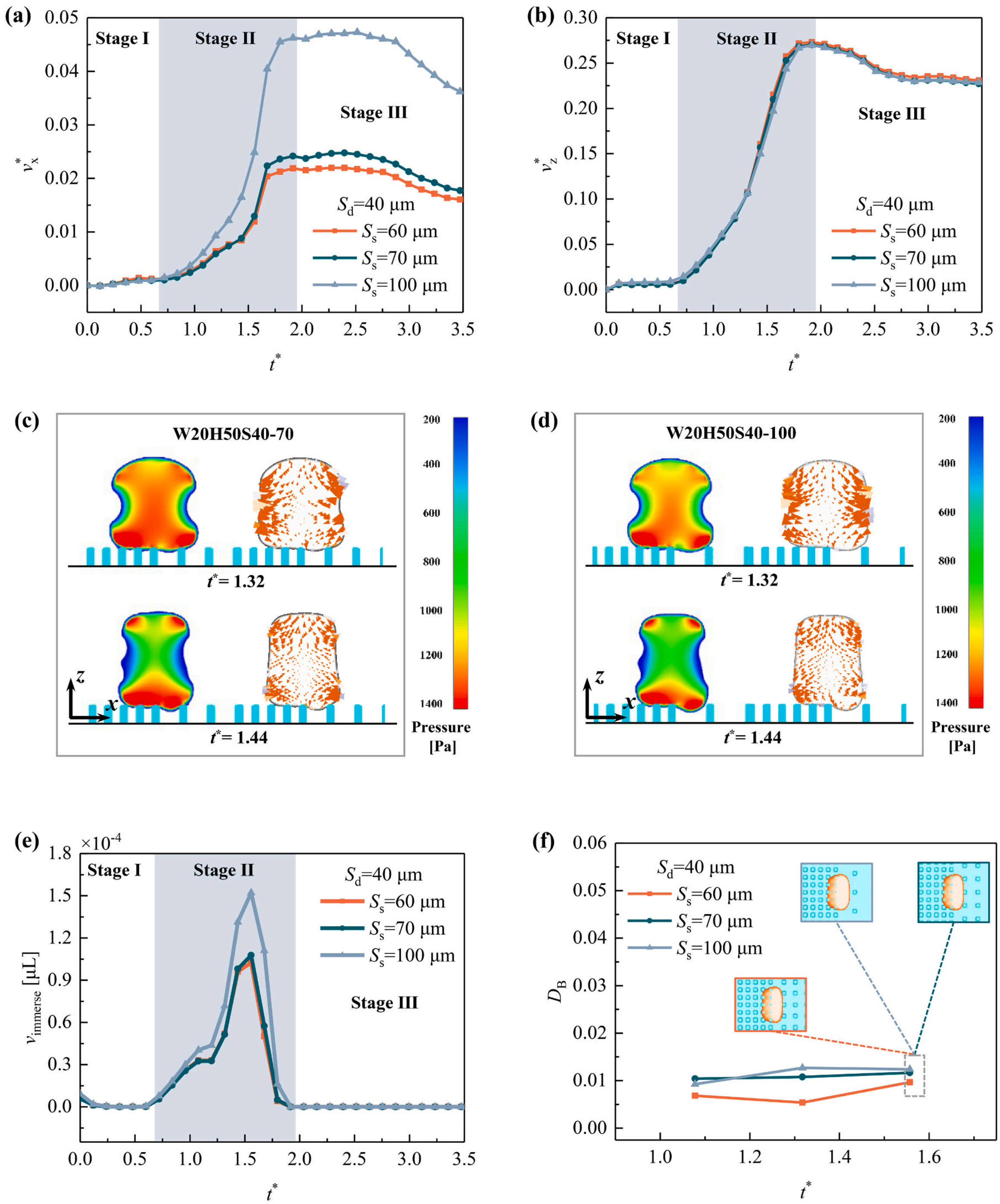

Fig. 7. Effect of micropillar spacing on droplet jumping on the non-uniform micropillar surface. (a)-(b) The variation of dimensionless center-of-mass velocity in the horizontal and vertical directions changes with time; (c)-(d) The pressure distribution and velocity vector diagrams inside the droplet on the surface of W20H50S40–70 and W20H50S40–100, respectively; (e) The variation of the liquid volume immersed into the micropillar gap with time; (f) The variation of the symmetry degree of the solid-liquid contact surface with time. On the three surfaces, the micropillar width is W= 20 μm, the micropillar height is H= 50 μm, and the micropillar spacing on the dense side is S_d= 40 μm.

(c) and (d), it could be observed that in the case of S_s = 100 μm, the left side of the droplet made contact with the micropillar at t^*=1.44, and there was adhesion between the solid and liquid, while the right side of the droplet detached from the micropillar, thus, leading to a more pronounced asymmetry in the distribution of high-pressure areas at the bottom of the droplet. Owing to the high-pressure area at the bottom serving as the propelling force for the droplet's horizontal motion, the horizontal velocity of the droplet would be greater when S_s = 100 μm. Upon comparing Fig. 7(a) and (e), it was evident that during the stage of liquid bridge shrinkage (Stage II), an increase in the micropillar spacing on the sparse side increased the liquid volume immersed in the micropillar gap and a subsequent rapid increase in horizontal velocity. These findings were in agreement with those presented in Fig. 5. Furthermore, it was evident from Fig. 7(f) that the effect of micropillar spacing on the asymmetry of the solid-liquid contact surface was negligible.

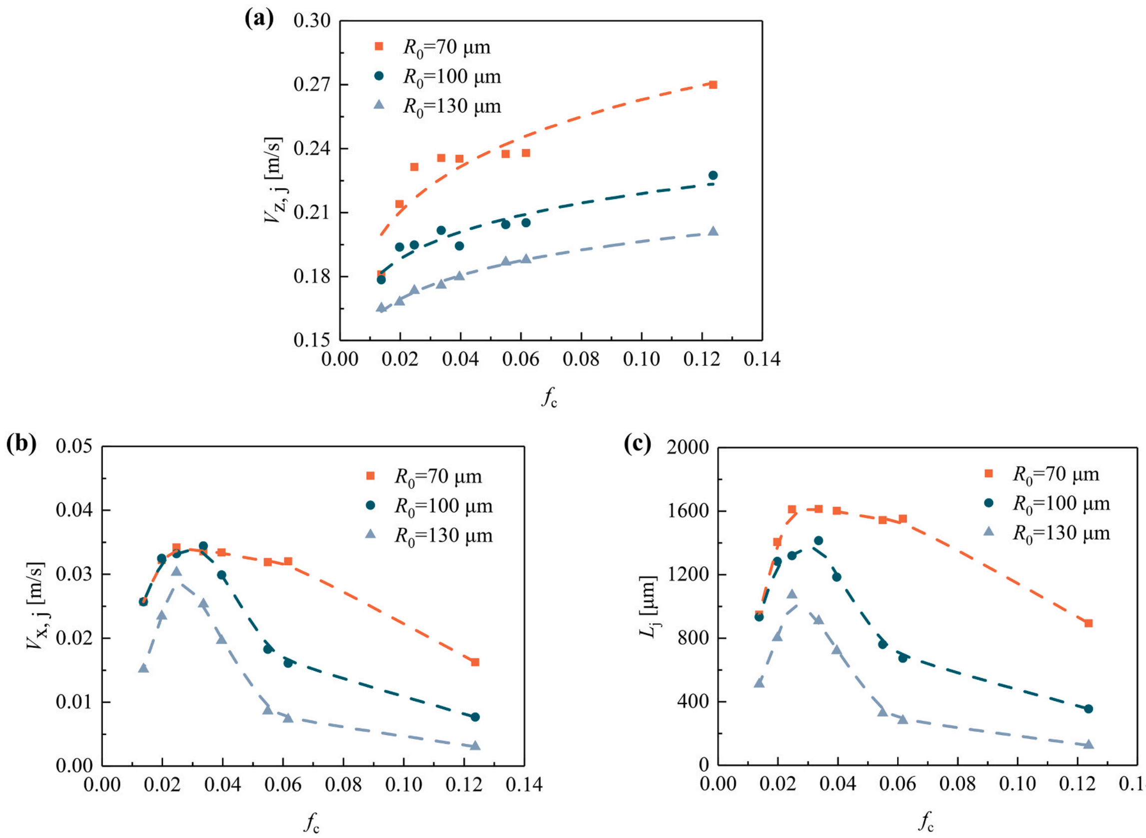

Fig. 8. On the non-uniform micropillar surfaces with different solid fractions, (a) the vertical jumping velocity of the droplet; (b) the horizontal jumping velocity of the droplet; (c) the horizontal transport distance of the droplet.

3.3. The solid fraction of non-uniform micropillar surface

Based on the preceding analysis, it was evident that in the case of non-uniform micropillars, changes in micropillar width resulted in variations in both vertical and horizontal velocity, while changes in micropillar spacing primarily impacted horizontal velocity. Modifying the micropillar dimensions by decreasing the width or increasing the spacing on the sparse side was more conducive to the movement of the droplets in the horizontal direction. Nevertheless, the influence of the micropillar height on the vertical and horizontal velocities could be neglected. To comprehensively examine the impact of micropillar width and spacing on droplet motion, based on the solid fraction of uniform micropillar surface[18–20,27], the solid fraction f_c of non-uniform micropillar surface was established:

$$f_c = \frac{W^2}{S_s^2}\left(1 - \frac{S_d^2}{S_s^2}\right), \ (S_d < S_s) \tag{10}$$

The larger the value of f_c was, the larger the micropillar distribution asymmetry between the dense and sparse sides was. Fig. 8 depicted the relationship between the solid fraction f_c and the vertical jumping velocity $V_{z, j}$, horizontal jumping velocity $V_{x, j}$, as well as the horizontal transport distance L_j of the droplet on a non-uniform micropillar surface. When the air resistance was not considered, the horizontal transport distance of the droplet was:

$$L_j = \frac{2V_{z, j}V_{x, j}}{g} \tag{11}$$

Fig. 8 illustrated that as the value of f_c increased, the vertical jumping velocity for droplets with the same radius also increased. However, the horizontal jumping velocity and the horizontal transport distance exhibited non-monotonic changes, initially increasing and subsequently decreasing. There was an optimal solid fraction to maximize the horizontal transport distance of droplets. Upon analyzing Fig. 8(a)-(c), it was apparent that there existed a significant correlation between the horizontal transport distance of the droplet and its horizontal jumping velocity. This observation suggested that the horizontal velocity was more crucial than the vertical velocity in droplet transport. Furthermore, the droplet's initial radius had an impact on both the vertical and horizontal jumping velocity, as well as the horizontal transport distance, all of which increased with the decrease of the initial radius of the droplet. It should be noted that when the droplet size was comparable to the structure size on a non-uniform micropillar surface, during the coalescence-induced jumping process, the bottom of the droplet immersed more into the micropillar gap on the sparse side, resulting in the asymmetry in the liquid bridge shrinkage and the high-pressure area distribution at the bottom of the droplet, which led to an unbalanced force in the horizontal direction. Eventually, the droplet's jumping direction was not perpendicular to the substrate. However, when the droplet radius was less than half of the micropillar spacing on the dense side, adjacent droplets could not touch and, therefore, could not coalesce and jump. And when the droplet radius was much larger than the micropillar, the droplet's horizontal velocity became very small, and the non-uniform distribution of micropillars could not transport the droplet in the horizontal direction (Section S6 of the supplementary material). Overall, through a systematic construction of the micropillar array and control of the relative size of droplets and micropillars, it was possible to adjust the jumping velocity and horizontal transport distance of droplets on superhydrophobic surfaces featuring non-uniform micropillars.

4. Conclusions

In this study, numerical simulation was used to examine the similarities and differences of coalescence-induced droplet jumping on superhydrophobic surfaces with uniformly and non-uniformly distributed micropillars, as well as the effect of micropillar width, height, and spacing on droplet jumping. On the sparse side of the non-uniform micropillar surface, due to the immersion of the lower portion of the droplet into the micropillar gap, the asymmetry of the liquid bridge contraction, the high-pressure area at the bottom of the droplet, and the morphology when the droplet jumped off the surface were caused, resulting in the decrease of the vertical velocity and the considerable increase of the horizontal velocity of the droplet. When the droplet radius remained constant, as the micropillar width increased, the horizontal velocity of the droplet decreased and the vertical velocity increased; as the micropillar spacing on the sparse side increased, the horizontal velocity increased, and the vertical velocity remained constant, but the change of the micropillar height had a negligible effect on the velocity in both directions. In addition, to analyze the coupling effect of micropillar width and spacing on coalescence-induced droplet jumping, the solid fraction of the non-uniform micropillar surface was proposed. By examining the process of droplet jumping on the non-uniform micropillar surface, the mechanism of the effect of surface structure on droplet jumping was disclosed in detail. This research clarified the droplet dynamics on the surface of various micropillar structures, and the findings could guide the design of superhydrophobic surface microstructures and have significant implications for specific applications of droplet jumping.

CRediT authorship contribution statement

Huimin Hou: Conceptualization, Methodology, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Xiaomin Wu:** Conceptualization, Writing – review & editing, Supervision, Project administration, Funding acquisition. **Zhifeng Hu:** Methodology, Formal analysis, Writing – review & editing. **Sihang Gao:** Methodology, Investigation, Writing – review & editing. **Liyu Dai:** Methodology, Writing – review & editing. **Zhiping Yuan:** Methodology, Investigation, Supervision, Writing – review & editing, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.colsurfa.2023.132127.

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